

Part II: Universal Measurable Estimators for ADAPT-VQE Gradient Selection

Steps 0–4 and Phases 2–4 — methods, results and reproduction

Part II working note

October 2026

Abstract

We implemented the Part II plan on the nine Part I benchmark cases and on states saved along exact ADAPT-VQE trajectories.

- **Step 0.** Each measurement context reads out a maximal group of 2^n commuting Pauli products. Measurement circuits, the moments of every group from one transform, and exact shot-level sampling are validated against dense linear algebra and against Part I.
- **Step 1.** The II-0 baseline runs Part I’s estimator as a shot-level experiment (pairwise elimination, $\gamma = 0.9$, top-up allocation). Its median cost is 62–83% below Part I’s M3 as run.
- **Steps 2–3.** 60–76% of $\mathcal{B}_0$ (mass completion) is measured by at least two contexts. With exact covariances, coefficient splitting (II-A) lowers the M3 cost by 38–80%. A design fitted to Hartree–Fock covariances costs 66–69% *more* than II-0 on correlated states.
- **Step 4.** Learned from the shots, with estimated radii, II-A costs 35–57% less than II-0 on the five fixed states run (H_4 CISD at two geometries, LiH, H_2O CISD at two geometries), with no wrong selections in 700 trials: 73–75% of the online oracle ceiling on H_4 and LiH, and as much as the oracle design on H_2O . (The H_4 and LiH numbers replace the 36/43/39% first reported, whose II-0 baselines predated the Step 4 fixes.)
- **Sign-aware elimination.** Part I’s pairwise rule plugs in estimated gradient signs. On a correlated ADAPT state of stretched H_4 it selected the wrong generator in 5–24% of trials for every configuration except II-A with oracle covariances, with heavy cost tails. A rule that uses pairwise contrasts only where signs are resolved selects correctly in every trial, at 21–31k shots. On fixed and trajectory states it costs -2 to $+16\%$ on H_2O , 6 – 38% on H_4 and 5 – 80% on LiH.
- **Contrast designs (II-E).** Designing the estimators of the decisive differences $s_{ig_l} - tg_i$ directly helps on the two hardest small-gap states (H_4 1.0 ADAPT5 and LiH ADAPT5: -32% and -36% against arm-wise II-A), is neutral on H_2O (-2 to $+6\%$), and hurts on wide-gap H_4 states ($+11$ to $+39\%$ against arm-wise II-A with the same rule).
- **Non-oracle start.** Starting from an a priori bound instead of $\max_i |g_i|$ changes the cost by -35 to $+11\%$ on most states, but it is unreliable on the states with the smallest gradients. On H_2O eq CISD ($|g_{(1)}| = 0.007$) the contrast design selects correctly in only 92% of trials, and on LiH ADAPT5 in 99% with a heavy tail: early decisions rest on near-deterministic contexts whose sample variance is zero.
- **Coverage.** With Part I’s convention (Bonferroni over the pool, a fresh look every round), 4–29% of trials have a round in which some interval misses the truth; selections stay correct because misses rarely involve the decisive pair. A union bound over rounds removes the misses at 1.8 – $2.6\times$ the cost where the variance estimates are sound, but not where small samples underestimate them (78% of trials on LiH ADAPT5 with the non-oracle start).
- **ADAPT trajectories (Phases 2 and 4).** Exact ADAPT reaches chemical accuracy in 10 (H_4 1.0), 6 (LiH), 17 and 18 (H_2O) steps; stretched H_4 stalls at 3.2 mHa. Every trajectory has exactly tied steps (spin partners), so selection must stop at a ρ -good leader. With measured selections ($\rho = 0.1$, fully non-oracle), every H_4 trajectory reproduces exact ADAPT and 88 of 90 LiH trajectories do (two take one extra step); all reach the exact endpoint. II-A lowers the cumulative selection cost by 41–44% against II-0 on H_4 and by 42% on LiH, and II-E by 60% on LiH. Tied steps dominate the cost.

All LiH and H_2O finite-shot runs were completed on the Trillium cluster and are included here.

Contents

1 Problem and conventions

2	Measurement model (Step 0)	2
3	Estimators and levels	3
4	Allocation, trajectories and cost metrics	3
5	Joint design solver (Step 3)	3
6	Online experiment (Steps 1 and 4)	4
6.1	Elimination rules and stopping	4
6.2	Contrast objective (U_{BAI} , II-E)	5
7	ADAPT trajectory states (Phase 2)	5
8	Phases 3 and 4: logged decisions and measured trajectories	5
9	Results	6
9.1	Steps 0–3 (unchanged)	6
9.2	Step 4: designs learned from the shots	6
9.3	Sign-aware rule, contrast objective and non-oracle start	7
9.4	Interval coverage	8
9.5	Phase 2: correlated ADAPT states	8
9.6	Phase 4: measured ADAPT trajectories	9
9.7	Cluster runs	10
10	Conclusions and next steps	10
11	Reproduction	11

1 Problem and conventions

At one ADAPT-VQE step, with state $|\psi\rangle$ and pool $\{G_i\}_{i=1}^N$, the task is to identify $i^* = \arg \max_i |g_i|$, where

$$g_i = \langle C_i \rangle_\psi, \quad C_i = [H, G_i] = \sum_\ell A_{i\ell} P_\ell,$$

using as few *context-shots* as possible. One context-shot is one state preparation plus one measurement in one fully commuting (FC) context.

Cases. The nine Part I cases, rebuilt with the current Part I pipeline:

- systems: H_4 square with side 1.0 and 2.0 Å; LiH at $R = 3.0$ Å; H_2O at $R_{\text{OH}} = 0.9572$ and 1.75 Å;
- states: HF and CISD (LiH HF only), plus the ADAPT trajectory states of Sec. 7;
- STO-3G, Jordan–Wigner, spin-orbital UCCSD pool ($N = 26, 92, 140$ on 8, 12, 14 qubits);
- Part I’s validation gate, and Part I’s deterministic first-fit FC partition of $\mathcal{B}_0 = \bigcup_i \text{supp}(C_i)$ into parent contexts $\mathcal{M}_\alpha$.

The rebuilt $|g_i|$ agree with Part I’s committed values to 3×10^{-10} . Some H_4 rows differ in the sign of some g_i ; this is an orbital-phase convention, and every quantity reported here is invariant under it.

Confidence and radii. A confidence radius is $r = z_\delta \epsilon$, with ϵ the one-standard-error target and $z_\delta = \Phi^{-1}(1 - \delta/2N)$, $\delta = 0.05$ (Bonferroni over the pool, as in Part I). The same z_δ is used for pairwise statements and in every round; Sec. 9.4 measures what this convention delivers. The gap is $|g_{(1)}| - |g_{(2)}|$, and $\Delta_i = |g_{(1)}| - |g_i|$.

Oracle versus measured. Steps 2–3 are oracle: exact covariances drive the designs and exact gradients fix the elimination trajectory. Their numbers are ceilings. Steps 1 and 4 and Phases 2–4 sample measurement outcomes; exact quantities enter a non-oracle run only through its validation record.

2 Measurement model (Step 0)

Measured groups. A context is measured by a Clifford U_α followed by readout of all n qubits. One shot therefore gives the eigenvalues of all 2^n elements of the maximal abelian group $\mathcal{G}_\alpha = \{\pm U_\alpha^\dagger Z^z U_\alpha\}$. A library Pauli P_ℓ is *measured by* α if $P_\ell \in \mathcal{G}_\alpha$, equivalently if it commutes with all n generators of $\mathcal{G}_\alpha$.

Completion. The parent members’ generators are completed to n by adding, one at a time, the first candidate that commutes with all current generators and lies outside their span. Candidate orders: *canonical* (single-qubit Z_q, X_q, Y_q) or *mass* (library Paulis by decreasing $\sum_i |A_{i\ell}|$); if a list runs out, a vector of the symplectic complement is used. A seeded *random* order is a control, and “span” (no completion) a diagnostic lower bound on overlap.

Circuit synthesis. From the tableau $[X|Z]$ of the n generators: row-reduce X and apply H on non-pivot qubits; row-reduce to $X = I$, which makes Z symmetric; apply S where $Z_{qq} = 1$ and CZ where $Z_{ab} = 1$; apply H on all qubits. Every Pauli of interest is then propagated through the gate list with exact phases, giving $U_\alpha P_\ell U_\alpha^\dagger = s_\ell Z^{z_\ell}$.

Moments and sampling. With outcome distribution $p_\alpha(b) = |\langle b|U_\alpha|\psi\rangle|^2$, all group means come from one Walsh–Hadamard transform:

$$E_\alpha[\mathbf{z}] = \sum_b p_\alpha(b)(-1)^{\mathbf{z} \cdot \mathbf{b}}, \quad \mu_\ell = s_\ell E_\alpha[\mathbf{z}_\ell], \quad \text{Cov}(P_\ell, P_m) = s_\ell s_m (E_\alpha[\mathbf{z}_\ell \oplus \mathbf{z}_m] - E_\alpha[\mathbf{z}_\ell]E_\alpha[\mathbf{z}_m]). \quad (1)$$

The exact p_α gives the oracle moments; a histogram of m sampled outcomes gives the empirical ones (factor $m/(m-1)$ on the covariance). Sampling m shots is one multinomial draw over 2^n outcomes.

3 Estimators and levels

For a fixed context library, every linear unbiased estimator has the form

$$\hat{g}_i = \sum_\alpha \mathbf{x}_{i\alpha}^\top \hat{\boldsymbol{\mu}}_\alpha, \quad \sum_{\alpha: P_\ell \in \mathcal{G}_\alpha} x_{i\alpha\ell} = A_{i\ell} \quad \forall \ell. \quad (2)$$

With fragments $F_{i\alpha} = \sum_\ell x_{i\alpha\ell} P_\ell$, coefficient matrix C and selection matrix B , Eq. (2) is exactly $A = BC$. Every design, including every learned design at every refit, is checked for $\max |A - BC| \leq 10^{-9} \max |A|$ and the run fails hard otherwise.

The levels differ only in which coordinates (α, ℓ) may be non-zero:

- II-0** $\ell \in \text{supp}(C_i)$ in its Part I parent context only: Part I’s estimator.
- II-A** $\ell \in \text{supp}(C_i)$ in every context that measures it (coefficient splitting).
- II-B** II-A plus Paulis with $A_{i\ell} = 0$ measured by at least two contexts already carrying C_i (zero-sum ghosts), capped at 200 per generator by a single-context control-variate score.
- II-E** Direct contrast design: the target row is $s_l A_l - t A_i$ for a pair (l, i) , over the union of both arms’ II-A coordinates. Pauli products whose coefficients cancel become zero-sum control variates (Sec. 6.2).

Coefficients are post-processing: any design satisfying Eq. (2) can be applied to every shot already taken.

4 Allocation, trajectories and cost metrics

Allocation. With n_α shots per context, $\text{Var } \hat{g}_i = \sum_\alpha q_{i\alpha}/n_\alpha$, where $q_{i\alpha} = \mathbf{x}_{i\alpha}^\top \Sigma_\alpha \mathbf{x}_{i\alpha}$. The allocation solves

$$\min_{\mathbf{n}} \sum_\alpha n_\alpha \quad \text{s.t.} \quad \text{Var } \hat{g}_i \leq \epsilon^2 \quad \forall i \in \mathcal{A}, \quad n_\alpha \geq c_\alpha. \quad (3)$$

With $c = 0$ this is Part I’s solver. With committed shots $c \neq 0$ (“top-up”) it is solved by multiplicative dual ascent, $n_\alpha = \max(c_\alpha, \sqrt{\sum_i \lambda_i q_{i\alpha}})$, $\lambda_i \leftarrow \lambda_i (\text{Var } \hat{g}_i / \epsilon^2)^{1/2}$, followed by a one-dimensional search for the smallest feasible scaling.

Cost metrics (Step 3).

M1 One allocation over all N arms at $r = \text{gap}/2$.

M3 bound Part I's exact-limit M3: at each breakpoint Δ_i , arms with $\Delta_i \leq 2r$ are active and resolved to $r = \Delta/2$.

M3 actual Part I's event-driven M3 on each arm's actual variance (primary metric).

Maxima accounting (Part I) solves each breakpoint from scratch and charges each context its largest request; *top-up* solves each breakpoint above the shots already committed. *Static* designs are optimised for the full pool; *redesign* (oracle II-D) re-optimises for the active set at every breakpoint.

5 Joint design solver (Step 3)

For an active set $\mathcal{A}$ the design problem $\min_{\mathbf{x}, \mathbf{n}} \sum_{\alpha} n_{\alpha}$ s.t. $\sum_{\alpha} q_{i\alpha}(\mathbf{x})/n_{\alpha} \leq \epsilon^2$ and Eq. (2) is jointly convex (quadratic-over-linear) and scale-free. Alternating exact per-generator designs with the allocation (iterative coefficient splitting) is monotone but stalls: a context whose shots reach zero can never be refilled (on the H₄ 1.0 HF winner it ends at 2.590 against an optimum of 2.3766, in units of ϵ^{-2}). Eliminating $\mathbf{n}$ gives the saddle problem

$$\min_{\mathbf{x}} \max_{\lambda \geq 0} \sum_{\alpha} h(Q_{\alpha}) - \epsilon^2 \sum_i \lambda_i, \quad Q_{\alpha} = \sum_i \lambda_i q_{i\alpha}, \quad h(Q) = n + Q/n, \quad n = \max(c_{\alpha}, \sqrt{Q}),$$

solved by an inner smoothed L-BFGS over all active arms' coefficients (smoothing $Q \rightarrow Q + \delta^2$, δ from 0.3 to 10^{-7} of the typical context norm) and an outer multiplicative dual ascent. The reported cost is the exact allocation of the returned design. It matches an independent single-arm minimisation to 10^{-4} and SLSQP on small joint instances to 2×10^{-4} ; on the large problems optimality is not certified.

6 Online experiment (Steps 1 and 4)

Loop. Part I's successive elimination: the radius starts at r_0 , shrinks as $r \leftarrow \gamma r$, and the run stops when one arm remains, when the leader is certified ρ -good (Sec. 6.1), or when r falls below a floor. Each round allocates shots for the active arms at $\epsilon = r/z$ (top-up), draws one multinomial histogram per context for the new shots, estimates $\hat{g}$, and eliminates. Every context that carries a fragment receives at least one shot (two in learned runs, one per fold). Trial i uses the random stream `SeedSequence(seed).spawn(n)[i]`, so results do not depend on how trials are split across processes or machines.

Starting radius. *Oracle start* (Part I): $r_0 = \max_i |g_i|$, floor $10^{-3}r_0$. *Bound start*: $r_0 = \max_{\ell} |A_{i\ell}|$, an a-priori bound on every $|g_i|$, floor $10^{-6}r_0$. With the bound start no exact quantity enters the run.

Learned designs (Step 4).

- *Covariance model.* Per context and fold, $\tilde{\Sigma} = \lambda \Sigma^{\text{prior}} + (1 - \lambda) \hat{\Sigma}^{\text{emp}}$, $\lambda = \nu/(\nu + m)$. Priors: the HF determinant, flat, none ($\nu = 0$), or the oracle (diagnostic).
- *Cross-fitting.* Every context's shots are split into two folds; the design fitted on one fold is applied to the other fold's means and the two estimates are averaged. Each half-estimate is unbiased whatever its design.
- *Minimum data.* A context carries learned coefficients only when both folds hold at least 50 shots in it.
- *Guard.* A fitted design is kept only if the other fold predicts a lower variance than the II-0 coefficients.
- *Schedule.* An II-0 pilot round, then refits whenever the shots held have grown by 1.5.
- *Radii and allocation* use only the held-out fold: $\text{Cov}(\hat{g}_i, \hat{g}_j) = \frac{1}{4} \sum_f \sum_{\alpha} \mathbf{x}_{i\alpha}^{f\top} \tilde{\Sigma}_{\alpha}^f \mathbf{x}_{j\alpha}^f / n_{\alpha}^f$, and shots are planned with exactly this per-shot variance.

Three failures of the first implementation motivated the last three safeguards: fitting on one or two samples per fold (54% correct on LiH), radii from a covariance including the fitting fold, and planning that disagreed with the radii (two H₄ 2.0 trials at 7.8×10^8 shots against a median of 1.3×10^4).

Small-sample radii. With the bound start, decisions can be taken when a context holds 2–5 shots per fold, where a sample covariance can be exactly zero. On a tied H₄ ADAPT state, II-0 with the bound start then stopped at 9,141 shots on an arm with 6% of the best gradient. All configurations introduced in this note therefore

replace the held-out covariance of a context with fewer than 50 shots by kI over its k Pauli products. This is a valid upper bound, since $\Sigma \preceq \text{tr}(\Sigma)I \preceq kI$ when every Pauli has variance at most one, and it is used for radii and planning alike, so the planner buys such contexts out of the small-sample regime. The Step 4 rows of Table 3 predate this rule.

6.1 Elimination rules and stopping

Let $r_i = z\sqrt{\text{Var } \hat{g}_i}$, $L_i = \max(0, |\hat{g}_i| - r_i)$, $U_i = |\hat{g}_i| + r_i$, and call the sign of g_k *resolved* when $|\hat{g}_k| > r_k$.

marginal Arm i is eliminated when $U_i < \max_j L_j$ (the work order’s absolute-interval rule; valid whatever the signs).

pairwise Part I’s Eq. (17): i goes when some j has $|\hat{g}_j| - |\hat{g}_i| > z\sqrt{\text{Var}(s_j\hat{g}_j - s_i\hat{g}_i)}$, with $s = \text{sign } \hat{g}$. The plug-in signs make it invalid while a sign is unresolved.

safe i goes when the marginal rule removes it, or when some j with a resolved sign satisfies

$$s_j\hat{g}_j - t\hat{g}_i > z\sqrt{\text{Var}(s_j\hat{g}_j - t\hat{g}_i)} \quad \text{for } t = s_i \text{ if } i \text{ is resolved, for both } t = \pm 1 \text{ otherwise.}$$

Given s_j , $|g_j| > |g_i|$ is equivalent to $s_jg_j > g_i$ and $s_jg_j > -g_i$, so the unresolved case needs no sign for i ; on the event that all intervals hold, no step is wrong.

A test reproduces the failure of the plug-in rule: with $g = (0.020, -0.010)$, standard deviations 0.05 and correlation -0.999 , pairwise eliminates the best arm in over 20% of draws ($z = 2$), the safe rule in under 5%.

ρ -good stopping. Exact BAI cannot separate an exact tie. With $\rho > 0$ the run stops, returning the leader l , once $s_l\hat{g}_l - (1 - \rho)t\hat{g}_i > z\sqrt{\text{Var}(\cdot)}$ for $t = \pm 1$ and every survivor i . Elimination stays exact, so on the good event the best arm survives and the returned arm has $|g_l| \geq (1 - \rho) \max_i |g_i|$.

Anytime validity. Optionally, round r uses $\delta_r = 6\delta/(\pi^2 r^2)$, which sums to δ over all rounds, so the family-wise statement covers the whole run instead of one round.

6.2 Contrast objective (U_{BAI} , II-E)

With the arm objective, each generator’s estimator minimises its own variance and the shots are planned so that every active arm reaches ϵ . At fixed shots the arms decouple, so this is also the minimiser of the work order’s $\mathcal{U}_{\text{avg}} = \sum_i w_i \text{Var } \hat{g}_i$ for any weights.

The contrast objective targets what elimination actually tests. Once the empirical leader’s sign is resolved, every other survivor i receives a directly designed, cross-fitted estimator of $s_l g_l - t g_i$ ($t = s_i$ if resolved, both $t = \pm 1$ otherwise), over the union of both arms’ II-A coordinates. Its starting point and guard fallback is the difference of the two arm designs. Shots are planned so that each contrast’s standard deviation reaches 2ϵ , the separation a pair of arms at radius ϵ would certify, and the leader’s own reaches ϵ . Designed contrasts take precedence over arm-implied ones in the safe rule. A test checks that contrast estimates are unbiased and that their held-out variances match the replicate spread (ratio 0.7–1.4). At II-0 the same construction changes only the allocation, which separates the allocation effect from the design effect.

7 ADAPT trajectory states (Phase 2)

Exact ADAPT-VQE is run from each HF case: $|\psi_k\rangle = e^{\theta_k G_{i_k}} \dots e^{\theta_1 G_{i_1}} |\text{HF}\rangle$, the next generator being the one of largest $|g_i|$ (ties to the lowest index), with all parameters re-optimised by BFGS on an exact adjoint gradient. Everything is done in the (N_α, N_β) sector (36, 225 and 441 determinants), where $g_i = 2 \text{Re}\langle H\psi | G_i \psi \rangle$. The run stops at $|E - E_{\text{FCI}}| < 1.6 \text{ mHa}$ or when every $|g_i| < 10^{-6}$.

Every state is saved and becomes a case `<geometry>_ADAPT<k>` for all scripts. Its gradient problem is built by Part I’s code from the *same* Hamiltonian object as the trajectory: a fresh SCF can return orbitals of opposite sign to the cached HF problem (seen on H_4), which would flip some gradient signs relative to the saved states. Gradients rebuilt from the commutator expansions agree with the sector computation to 10^{-8} at every step.

On stretched H_4 every gradient vanishes by symmetry after 10 steps, 3.2 mHa above FCI: a property of the spin-orbital pool, not of the measurement. Snapshots for fixed-state benchmarks were chosen mid-trajectory with

Table 1: Exact ADAPT trajectories. *Ties*: steps whose two leading $|g_i|$ are equal (spin partners). *Small gaps*: steps with relative gap below 5%.

Geometry	steps	final $ E - E_{\text{FCI}} $	ties at step	small gaps at step
H ₄ side 1.0	10	1.09 mHa	2, 4, 8	1, 2, 4, 6, 8
H ₄ side 2.0	10	3.22 mHa (stalls)	1, 4, 8	1, 4, 6, 8
LiH $R = 3.0$	6	0.68 mHa	1, 4	1, 2, 4
H ₂ O eq	17	1.54 mHa	2, 4, 8	2, 3, 4, 6, 8, 9, 12, 14
H ₂ O stretched	18	1.43 mHa	2, 4	2, 4, 5, 7, 12, 13, 15, 16

distinct leaders (H₄ 1.0 ADAPT3 and ADAPT5, H₄ 2.0 ADAPT2 and ADAPT5, LiH ADAPT3 and ADAPT5, H₂O eq ADAPT11, H₂O stretched ADAPT8), plus tied states for ρ -good stopping.

8 Phases 3 and 4: logged decisions and measured trajectories

Phase 3 record. `scripts/phase3_selection_record.py` runs complete non-oracle decisions with a run log and writes the work order’s required outputs: (i) A , labels and Hamiltonian; (ii) the library, multiplicities and auxiliary rule; (iii) every context with its Clifford circuit and images; (iv) the fold designs of every refit as sparse C rows; (v) the $A = BC$ check at every refit (hard failure); (vi) outcome histograms per context and fold (the sufficient statistics); (vii) fold-model and prior covariance matrices of the most-used contexts; (viii)–(ix) per round, the shots added per context and fold, estimates, intervals, resolved signs, leader, eliminations with the arm and rule responsible, and designed contrasts; (x) context-shots, two-qubit gate counts and classical time; (xi) exact gradients and interval coverage in a separate `*_validation.json`. Six logged decisions on LiH ADAPT3 (three per configuration, non-oracle start) were all correct, with no interval missing the truth in any round. II-0 (safe rule) took $1.09\text{--}1.27 \times 10^6$ shots in 47–49 rounds, and II-E took $4.8\text{--}6.8 \times 10^5$ shots in 52–53 rounds, with 13 refits and 6–10 min of design time per decision.

Phase 4 driver. `scripts/phase4_adapt_selection.py` grows the ansatz with generators chosen by a Part II method from sampled outcomes on the current state; parameters are re-optimised exactly. Only selection shots are counted; energy estimation is outside Part II. Every method is fully non-oracle inside a selection (estimated radii, bound start, safe rule, small-sample radii) and stops at a ρ -good leader with $\rho = 0.1$. The driver’s own stopping rule (energy within 1.6 mHa, or all exact $|g_i| < 10^{-6}$, or 10 steps beyond exact ADAPT) is not measured.

9 Results

9.1 Steps 0–3 (unchanged)

- *Step 0.* Circuits, group moments and the sampler are validated (sampler bias below 3.6σ over 2,000/400 replicates, variance and correlation errors within $5\sqrt{2/R}$ and $5/\sqrt{R}$, down to 86 shots in total).
- *Step 1.* The online loop reproduces Part I’s noiseless trial exactly (real-valued shots) and its Gaussian surrogate within 1.5 standard errors. II-0 (pairwise rule, $\gamma = 0.9$, top-up, oracle radii) has median cost 285,250 / 39,754 / 160,459 / 22,136 / 299,479 on H₄ 1.0 HF / CISD, H₄ 2.0 HF / CISD and LiH, 62–83% below Part I’s M3 as run, with 5 wrong selections in 1,100 trials, all on the tied H₄ HF rows.
- *Step 2.* Mass completion: multiplicity 1.95 (H₄), 2.95 (LiH), 3.10 (H₂O); 60–76% of $\mathcal{B}_0$ splittable, carrying 62–89% of the coefficient mass.
- *Step 3.* On M3 actual, II-A lowers the cost by 38–54% on H₄, 73–76% on LiH and 70–80% on H₂O (Table 2). II-B adds 2–10 points on H₄ and nothing resolvable on LiH and H₂O. Redesign at every breakpoint adds 17–19 points on the depth-dominated H₄ HF rows and 15–16 points on stretched H₂O HF, where about 63% of the static design’s cost falls after the first elimination; it adds 1–3 points on LiH and the other H₂O states (11–25% after the first elimination). It helps only with top-up accounting. HF-fitted designs cost 66–69% more than II-0 on the CISD states.

The other Step 0–3 tables are in `part2_estimator_design/RESULTS.md` and in the previous version of this note.

Table 2: Oracle ceilings (Step 3), mass completion. II-0: M3 on actual precision, context-shots. II-A, II-B: change against II-0 on the same metric. Redesign: oracle II-D at every elimination breakpoint, change against II-0 on the M3 bound, both with top-up accounting. The LiH and H₂O rows are the Oct 2 runs (`runs/step3_oracle_ceiling_table5.csv`). The solver minimises the M3 bound, not M3 actual: the earlier solve of II-A on those five states reached the bound within 2 points but M3 actual within 11 points (−70%, −80%, −75%, −64%, −68%), so II-A/II-B differences of that size are not significant.

Case	II-0	II-A	II-B	redesign (II-A / II-B)
H ₄ 1.0 HF	569,020	−38%	−40%	−57% / −58%
H ₄ 1.0 CISD	49,729	−54%	−56%	−53% / −56%
H ₄ 2.0 HF	228,733	−40%	−50%	−59% / −60%
H ₄ 2.0 CISD	31,553	−43%	−50%	−50% / −57%
LiH HF	333,272	−73%	−71%	−83% / −83%
H ₂ O eq HF	1.32×10^7	−80%	−83%	−83% / −83%
H ₂ O eq CISD	2.44×10^{10}	−76%	−78%	−80% / −81%
H ₂ O str HF	1.46×10^7	−70%	−68%	−88% / −88%
H ₂ O str CISD	1.22×10^8	−79%	−69%	−75% / −75%

9.2 Step 4: designs learned from the shots

Table 3: Learned designs, pairwise rule, oracle start. Mean (change against II-0 with the same kind of radii) and correct selections, out of 200 on H₄ and 100 on LiH. The three baseline rows were rerun after the Step 4 fixes; the first version of this note compared the II-A rows with baselines run before them.

Design	radii	H ₄ 1.0 CISD		H ₄ 2.0 CISD		LiH 3.0 HF	
		mean (change)	ok	mean (change)	ok	mean (change)	ok
II-0	oracle	43,864	200	26,751	200	307,885	100
II-A, oracle covariance	oracle	24,317 (−45%)	200	11,643 (−56%)	200	146,538 (−52%)	100
II-A, HF prior only	oracle	1.19×10^9	200	7.39×10^8	194	—	—
II-0	estimated	41,463	200	24,699	200	304,990	100
II-A, HF prior only ($\nu = \infty$)	estimated	81,047 (+95%)	197	143,851 (+482%)	171	—	—
II-A, HF shrunk, $\nu = 1000$	estimated	32,099 (−23%)	200	31,696 (+28%)	195	—	—
II-A, HF shrunk, $\nu = 100$	estimated	28,307 (−32%)	200	15,293 (−38%)	200	—	—
II-A, flat prior, $\nu = 100$	estimated	—	—	—	—	193,042 (−37%)	100
II-A, data only ($\nu = 0$)	estimated	27,129 (−35%)	200	13,727 (−44%)	200	184,511 (−40%)	100
II-A, data only, no guard	estimated	26,749 (−35%)	200	13,764 (−44%)	200	184,511 (−40%)	100

- *Learned from the data, II-A keeps 73–75% of its ceiling.* It costs 35%, 44% and 40% less than II-0 run the same way; the online oracle ceiling is 45%, 56% and 52%. All 500 trials selected the winner, and the data-only rows have no tail.
- *The same holds on H₂O* (Table 4): learned II-A costs 57% and 54% less than II-0 on the equilibrium and stretched CISD states, as much as the design with exact covariances, with every selection correct.
- *Estimated radii are nearly free at II-0* (within 6% of oracle radii).
- *The HF prior alone fails*, in the design and through near-zero variances in the radii and allocation. A prior helps only with little weight and never beat the data alone; the guard rarely acts once the minimum-data rule is in place.

9.3 Sign-aware rule, contrast objective and non-oracle start

- *The bound start is cheap but not always safe.* Removing the last oracle input (the starting radius) changes the cost by −35 to +11% at matched rule and objective on most states. On H₂O eq CISD, the state with the smallest gradients ($|g_{(1)}| = 0.007$, gap 0.002), it selects correctly in 99% (II-0) and 92% (II-E) of trials, with intervals missing the truth in 76% and 99% of trials. The early, coarse rounds take decisions on contexts with a few tens of shots whose rare outcomes have not yet appeared, so their sample variance is zero; the kI bound only covers contexts below 50 shots per fold.
- *The safe rule costs +6 to +13% at II-0 and +21 to +38% at II-A on H₄, 0 and +16% on H₂O CISD, and*

Table 4: H₂O CISD states, 100 trials each (two shards on Trillium), estimated radii unless marked. Mean context-shots (change against II-0, pairwise) and, where not 100, the percentage of correct selections in brackets.

Configuration	H ₂ O eq CISD	H ₂ O stretched CISD
II-0, pairwise	2.05×10^{10}	8.63×10^7
II-0, pairwise, oracle radii	2.07×10^{10} (+1%)	9.19×10^7 (+6%)
II-A oracle covariance, pairwise	9.90×10^9 (−52%)	3.99×10^7 (−54%)
II-A HF $\nu = 100$, pairwise	8.78×10^9 (−57%)	3.92×10^7 (−55%)
II-A data, pairwise	8.80×10^9 (−57%)	3.95×10^7 (−54%)
II-0, safe	2.05×10^{10} (+0%)	9.98×10^7 (+16%)
II-0, safe, bound start	2.02×10^{10} (−2%) [99]	1.02×10^8 (+18%)
II-A data, contrast	8.65×10^9 (−58%)	3.92×10^7 (−55%)
II-A data, contrast, bound start	2.31×10^9 (−89%) [92]	2.55×10^7 (−70%)

Table 5: Fixed states, all configurations with estimated radii. Mean context-shots (change against II-0 with the pairwise rule and estimated radii, Table 3); every row selected the winner in every trial. “Bound”: non-oracle start. “Anytime”: union bound over rounds. LiH: 100 trials. H₂O: Table 4.

Configuration	H ₄ 1.0 CISD	H ₄ 2.0 CISD	LiH 3.0 HF
II-0, pairwise (reference)	41,463	24,699	304,990
II-0, safe	47,037 (+13%)	26,126 (+6%)	549,094 (+80%)
II-0, safe, bound start	42,411 (+2%)	24,815 (+0%)	568,140 (+86%)
II-0, contrast allocation	61,080 (+47%)	35,031 (+42%)	–
II-A data, pairwise	27,129 (−35%)	13,727 (−44%)	184,511 (−40%)
II-A data, safe	32,814 (−21%)	18,962 (−23%)	–
II-A data, contrast	45,569 (+10%)	21,114 (−15%)	238,484 (−22%)
II-A data, contrast, bound start	40,618 (−2%)	17,926 (−27%)	227,290 (−25%)
II-A data, contrast, bound start, anytime	74,255 (+79%)	38,420 (+56%)	–
II-A oracle covariance, pairwise	24,317	11,643	146,538
II-A oracle covariance, contrast	37,027	15,297	146,205

80% at II-0 on LiH. The safe rows also carry the small-sample radius bound, so the two effects are not separated; on LiH many lightly sampled contexts sit under the conservative kI bound for longer.

- *The contrast objective does not pay on these fixed states.* On H₄ 1.0 CISD it is 39% dearer than arm-wise II-A with the same rule, and even with oracle covariances it is 52% dearer than the arm objective. Contrast planning charges the leader’s own ϵ plus a 2ϵ contrast for every survivor; when the gap is large, resolving each arm to ϵ is cheaper. On LiH the two objectives tie at the oracle level, and on H₂O CISD the learned contrast design is within 2% of learned II-A.

9.4 Interval coverage

The validation record counts, per trial, whether any interval used in any round (marginal for every active arm, and every designed contrast) missed the exact value.

Table 6: Share of trials with at least one interval missing the truth in some round. Nominal family-wise level per round: $\delta = 0.05$.

Configuration	H ₄ 1.0 CISD	H ₄ 2.0 CISD	LiH 3.0 HF
II-0, pairwise, oracle radii	0.22	0.25	0.14
II-0, pairwise, estimated radii	0.23	0.29	0.22
II-A, oracle covariance	0.12	0.19	0.08
II-0, safe	0.16	0.04	–
II-A data, safe	0.09	0.04	–
II-A data, contrast, bound start	0.12	0.14	–
II-A data, contrast, bound start, anytime	0.00	0.00	–

Part I’s convention does not deliver family-wise coverage over a whole run: a fresh look every round with

a per-round Bonferroni bound misses in 4–29% of trials (19–21% for II-0 on H₂O CISD). The selections are nevertheless correct, because misses rarely involve the decisive pair. The anytime bound removes every miss on H₄ and LiH ADAPT3, at 1.8–2.6× the cost (Tables 5, 7 and 8). It does not on LiH ADAPT5 with the bound start (78% of trials still miss): there the misses come from underestimated variances, not from repeated looks. On the trajectory state H₄ 2.0 ADAPT2 the learned pairwise configurations miss in 63–78% of trials and the learned safe configurations in 1–2%.

9.5 Phase 2: correlated ADAPT states

Table 7: ADAPT snapshots of H₄, 100 trials each, estimated radii unless marked. Mean context-shots and, where not 100, the percentage of correct selections in brackets. [†]Tail-dominated mean; medians: H₄ 1.0 ADAPT5 II-0 oracle radii 558,631; H₄ 2.0 ADAPT2 II-0 26,920, II-0 oracle radii 9.05×10^7 , II-A data 17,770, HF prior 5,610, HF $\nu = 100$ 7,807.

Configuration	H ₄ side 1.0		H ₄ side 2.0	
	ADAPT3 (gap 33%)	ADAPT5 (gap 15%)	ADAPT2 (gap 11%)	ADAPT5 (gap 56%)
II-0, pairwise	26,504	411,200	789,821 [†] [95]	58,402
II-0, pairwise, oracle radii	26,455	2.92×10^8 [†] [99]	6.70×10^7 [†] [92]	53,636
II-A oracle covariance, pairwise	14,240	239,815	16,622	26,895
II-A data, pairwise	18,161	288,503 [99]	246,207 [†] [95]	29,749
II-A HF prior only, pairwise	16,954	455,298 [99]	39,181 [†] [76]	69,934
II-A HF $\nu = 100$, pairwise	17,751	246,442	80,193 [†] [85]	32,520
II-0, safe	29,883	406,671	28,031	55,818
II-0, safe, bound start	28,233	399,827	30,726	53,882
II-A data, contrast	27,787	197,352	23,357	34,881
II-A data, contrast, bound start	24,513	218,787 [99]	24,023	30,269
II-A data, contrast, bound start, anytime	51,251	568,295	62,195	62,772
II-A oracle covariance, contrast	17,895	193,558	21,049	28,543

- *The plug-in sign rule fails on correlated trajectory states.* On H₄ 2.0 ADAPT2 every pairwise configuration except II-A with oracle covariances made wrong selections (II-0 95%, II-0 with *oracle* radii 92% and a 6.7×10^7 mean, II-A data 95%, HF prior 76%). With the safe rule every configuration selected correctly in every trial, at 21–31k shots (62k with the anytime bound). Near-zero gradients with unresolved signs and strong correlations are exactly the case the plug-in signs get wrong.
- *On hard states the contrast objective is the best non-oracle method.* On H₄ 1.0 ADAPT5 (gap 15%) it costs 197,352, 32% below arm-wise II-A data and 52% below II-0, and matches the oracle contrast design.
- *Splitting still pays on trajectory states.* With the safe rule and the bound start, II-A contrast costs 13–45% less than II-0.
- *The HF prior is unreliable along the trajectory:* the cheapest non-oracle row on ADAPT3, but 76% correct on ADAPT2.

Tied states. On H₄ 1.0 ADAPT4 (exact tie) exact BAI ran to 1.66×10^9 shots in a single trial. With $\rho = 0.1$, II-0 safe (bound start) stops at a mean of 493,275 and II-A contrast at 236,115 shots; on H₄ 2.0 ADAPT1, 24,633 and 17,361. Every selection was ρ -good: 36–52% picked the lower-indexed generator of the tie, 44–54% its tie partner, and 4–10% a third generator within 3–5% of the best.

LiH and H₂O snapshots (Table 8).

- Learned splitting pays on every state: II-A data costs 34–39% less than II-0 on LiH and 53–58% less on H₂O, with every selection correct.
- The contrast design is the best non-oracle method on LiH ADAPT5 (gap 15%), 36% below II-A data and level with its oracle form. On H₂O it is within +1 to +6% of II-A, and on the wide-gap LiH ADAPT3 it is 39% dearer.
- The HF prior alone selects correctly in only 29% of trials on LiH ADAPT5. Shrunk to the data ($\nu = 100$), it matches the data alone within 8% on all four states.
- The bound start makes the contrast design 13–19% cheaper on H₂O, but on LiH ADAPT5 it costs a 1% error rate and a heavy tail (median 5.4×10^7 , mean 1.2×10^8).

Table 8: ADAPT snapshots of LiH and H₂O, 100 trials each, estimated radii unless marked. Mean context-shots (change against II-0, pairwise) and, where not 100, the percentage of correct selections in brackets. “–”: not run. Relative gaps: LiH ADAPT3 75%, ADAPT5 15%, H₂O eq ADAPT11 11%, stretched ADAPT8 15%.

Configuration	LiH ADAPT3	LiH ADAPT5	H ₂ O eq ADAPT11	H ₂ O str ADAPT8
II-0, pairwise	709,261	6.60×10^7	1.93×10^8	1.84×10^8
II-0, pairwise, oracle radii	707,341 (−0%)	6.60×10^7 (+0%)	–	–
II-A oracle covariance, pairwise	375,474 (−47%)	3.61×10^7 (−45%)	7.43×10^7 (−62%)	9.03×10^7 (−51%)
II-A data, pairwise	430,494 (−39%)	4.34×10^7 (−34%)	8.17×10^7 (−58%)	8.57×10^7 (−53%)
II-A HF prior only, pairwise	437,864 (−38%)	2.36×10^7 (−64%) [29]	–	–
II-A HF $\nu = 100$, pairwise	430,152 (−39%)	4.14×10^7 (−37%)	7.56×10^7 (−61%)	8.61×10^7 (−53%)
II-0, safe	1.14×10^6 (+61%)	6.92×10^7 (+5%)	2.01×10^8 (+4%)	1.81×10^8 (−2%)
II-0, safe, bound start	1.21×10^6 (+71%)	6.64×10^7 (+1%)	1.92×10^8 (−1%)	1.85×10^8 (+1%)
II-A data, contrast	600,144 (−15%)	2.77×10^7 (−58%)	8.24×10^7 (−57%)	9.10×10^7 (−50%)
II-A data, contrast, bound start	530,004 (−25%)	1.16×10^8 (+76%) [99]	6.66×10^7 (−66%)	7.88×10^7 (−57%)
II-A data, contrast, bound start, anytime	910,289 (+28%)	1.93×10^8 (+192%)	–	–
II-A oracle covariance, contrast	370,494 (−48%)	2.91×10^7 (−56%)	7.81×10^7 (−60%)	9.44×10^7 (−49%)

- On the tied LiH ADAPT4, $\rho = 0.1$ stopping costs 4.1×10^7 (II-0) and 1.6×10^7 shots (II-E, −62%). 100% and 97% of the selections are ρ -good, so II-E’s 3% exceeds δ , for the same small-sample reason.

9.6 Phase 4: measured ADAPT trajectories

Table 9: Measured ADAPT trajectories (48 per method on H₄, 30 on LiH), $\rho = 0.1$, fully non-oracle selections. Cumulative selection context-shots over the whole trajectory. Every trajectory ended at the exact-ADAPT energy; all H₄ and 88 of 90 LiH trajectories took the same number of steps. “Best”: selections whose $|g|$ equals the maximum (ties included).

Geometry	method	median	P90	best	ρ -good
H ₄ side 1.0 (10 steps, 1.09 mHa)	II-0, safe	1.88×10^6	2.34×10^6	93%	100%
	II-A data, safe	1.11×10^6 (−41%)	1.44×10^6	92%	100%
	II-A data, contrast	1.09×10^6 (−42%)	1.61×10^6	90%	100%
H ₄ side 2.0 (10 steps, stalls at 3.22 mHa)	II-0, safe	1.59×10^7	2.29×10^7	94%	100%
	II-A data, safe	9.39×10^6 (−41%)	1.56×10^7	93%	100%
	II-A data, contrast	8.85×10^6 (−44%)	1.56×10^7	94%	100%
LiH (6 steps, 0.68 mHa)	II-0, safe	8.57×10^7	1.16×10^8	94%	100%
	II-A data, safe	4.94×10^7 (−42%)	7.66×10^7	94%	99.4%
	II-A data, contrast	3.41×10^7 (−60%)	5.52×10^7	96%	99.4%

- Measured selections reproduce exact ADAPT: same length and final energy in all 288 H₄ trajectories; the mean shortfall of the selected $|g|$ is 0.13–0.20%. On LiH one selection per learned method (of 181) was not ρ -good (shortfall 13–15%); each such trajectory took one extra step and still reached the same energy.
- II-A lowers the cumulative selection cost by 41–44% against II-0 at the same rule and stopping, and II-E by 42–60%, most on LiH.
- The cost is dominated by tied and small-gap steps: on H₄ 2.0 the tied step 8 ($\max |g_i| = 0.0094$) takes $7\text{--}14 \times 10^6$ of the total, and on H₄ 1.0 the tied steps 4 and 8 take about half. Without ρ -good stopping these steps would not terminate.

9.7 Cluster runs

The LiH snapshots, the tied ADAPT4 state, LiH Phase 4 and Phase 3, and all H₂O runs ran on Trillium: two LiH node-jobs (2.2 and 5.1 h) and eight H₂O node-jobs (11–17.5 h, 60 workers each, memory-bound), about 110 node-hours in total. None needed a requeue. On H₂O one II-A trial costs about 6 min of CPU time, II-0 with the bound start 15 min, and a learned contrast trial 1.4–5.5 h of design time.

10 Conclusions and next steps

1. The shot-level II-0 baseline is 62–83% cheaper than Part I’s M3 as run, and coefficient splitting has a large oracle ceiling (38–80% on M3 actual).
2. Learned from the shots, II-A costs 35–57% less than II-0 without oracle input (H_4 CISD, LiH, H_2O CISD) and 30–58% less on ADAPT trajectory states, provided it uses cross-fitting, held-out radii and allocation, and a minimum number of shots per fold. That is 73–75% of the oracle ceiling on H_4 and LiH and all of it on H_2O . Data-driven covariances should be the default; the HF prior should never be used alone (29% correct on LiH ADAPT5).
3. Part I’s plug-in sign rule is not safe on correlated ADAPT states (5–24% wrong selections on H_4 2.0 ADAPT2). The sign-aware rule fixes this at -2 to $+16\%$ extra cost on H_2O and 6–38% on H_4 , but up to 80% on LiH; separating the rule from the small-sample radius bound on LiH is the next measurement.
4. The starting-radius oracle can be removed on most states at no cost, but not yet safely on the hardest ones (92% correct on H_2O eq CISD): variance estimates from a few tens of shots of near-deterministic contexts need a better upper bound (for example a Bernoulli upper confidence bound) before the non-oracle start can be the default.
5. Direct contrast design (II-E) is the method of choice on hard, small-gap states (-32% and -36% against II-A; -60% against II-0 over LiH trajectories), neutral on H_2O , and should not be used when the gap is large. A natural rule is to switch from arm to contrast planning once few arms survive.
6. Per-round Bonferroni does not give run-wide coverage. Report it as such, or pay $1.8\text{--}2.6\times$ for the anytime bound, which is only as good as the variance estimates it is applied to.
7. ADAPT trajectories contain exact ties at every geometry; ρ -good stopping is required, and with it measured selection reproduces exact ADAPT. Ties dominate the cumulative cost, so the next gains are in how quickly a tie is certified ρ -good.

Next. Make small-sample variance bounds valid for near-deterministic contexts; run measured trajectories on H_2O ; decompose the LiH cost of the safe rule; switch between arm and contrast objectives adaptively; replace per-generator refits by the joint solver; price circuits (8–32 CZ per context) and classical design time (now logged per run).

11 Reproduction

Code. `part2_estimator.design/` on branch `part2`; it imports Part I’s `src/` directly. New modules: `rules` (elimination and stopping), `trajectory` (ADAPT states), `runlog` (Phase 3 record); `learning` gained the contrast objective, the bound start, the small-sample radius bound and anytime validity; `parallel` gained checkpoints.

Environment. Local: WSL Ubuntu, Python 3.12.3, NumPy 2.5.2, SciPy 1.18.1, PySCF 2.14.0, OpenFermion 1.8.1 (`requirements.txt`). Trillium: Python 3.12.4 with NumPy 2.4.2, SciPy 1.17.1, PySCF 2.14.0, OpenFermion 1.8.1 from the Alliance wheelhouse. Per-trial seeding makes results independent of process layout; they are statistically, not bitwise, comparable across library versions.

Commands (from `part2_estimator.design/`; `OMP_NUM_THREADS=1` with `--workers`):

```
python -m pytest tests -q # 67 tests
python scripts/phase2_adapt_states.py --cases H4_square_eq_side1p0_HF \
    H4_square_stretch_side2p0_HF LiH_R3p0_HF H2O_eq_HF H2O_stretch_HF
python scripts/step4_learned_designs.py --cases H4_square_eq_side1p0_CISD \
    --configs "II-0 safe" "II-A data, contrast" ... --trials 200 --workers 12
python scripts/step4_learned_designs.py --cases H4_square_eq_side1p0_ADAPT5 \
    --configs ... --trials 100 --workers 12
python scripts/phase3_selection_record.py --cases LiH_R3p0_ADAPT3 --trials 3
python scripts/phase4_adapt_selection.py --cases H4_square_eq_side1p0_HF \
    --trajectories 48 --workers 12
python scripts/merge_trials.py --step step4 --cases <cases>
python scripts/aggregate_results.py
```

The Step 0–4 commands of the previous version are unchanged. Configuration names are the keys of `CONFIGS` in `scripts/step4_learned_designs.py`.

Cluster. `cluster/trillium_run.sbatch` runs one whole node per job file in `cluster/jobs/`; the lines of a job file start together. Every finished trial and trajectory is checkpointed; with `--resume` a rerun skips finished configurations and trials, and the job requeues itself 15 min before the 24h limit (at most twice). A resumed run reproduces an uninterrupted one exactly (tested). H₂O jobs are memory-bound at 60 workers per node; LiH jobs use about 190 workers.

Outputs. Each case writes `runs/<case>/<case>_<step>.csv` (rebuilt from all per-trial files, so runs of different configurations accumulate), `runs/<case>/trials/`, `runs/<case>/phase3/` and `runs/<case>/phase4/`, each with a `*meta.json` recording package versions and the git commit. Trajectory states are in `.cache/trajectory/`. Seeds: Step 4 and Phase 2 snapshots 2, Phase 4 7, Phase 3 5.